

MITRE ATT&CK; Research Internship Report

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Intern ID: 2184

Assigned Campaign: 84

Software Range: 840–850

Research Area: APT Group Research, Threat Intelligence & Monitoring

Framework: MITRE ATT&CK; Framework

This report was prepared according to the MITRE ATT&CK; Research Internship instructions. The objective is to demonstrate understanding of cyber threat intelligence, Advanced Persistent Threat (APT) groups, malware campaigns, Indicators of Compromise (IoCs), MITRE ATT&CK; techniques, and defensive monitoring strategies.

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1. Introduction to Threat Intelligence

Threat intelligence is the collection, analysis, and sharing of information related to cyber threats, attackers, malware, phishing infrastructure, ransomware groups, and suspicious cyber activities. Organizations use threat intelligence to improve detection, prevention, and response capabilities.

2. Advanced Persistent Threat (APT) Groups

Advanced Persistent Threat (APT) groups are organized threat actors conducting long-term cyber operations. These groups are usually associated with espionage, financial theft, sabotage, and infrastructure attacks. Commonly researched groups include Lazarus Group, APT29, Sandworm Team, and Gallmaker.

3. MITRE ATT&CK; Framework

MITRE ATT&CK; is a globally recognized cybersecurity knowledge base used for mapping attacker behavior. It contains tactics, techniques, and procedures (TTPs) observed in real-world attacks. Security teams use ATT&CK; for detection engineering, threat hunting, malware analysis, and incident response.

4. Threat Actor Research

Threat actor research includes studying attacker behavior, attack timelines, malware usage, targeted industries, phishing campaigns, infrastructure abuse, and command-and-control operations. Threat intelligence analysts also monitor open-source intelligence (OSINT) platforms and leak sites.

5. Malware Monitoring & Analysis

The following malware families and tools were monitored during this research:

- Agent Tesla
- Cobalt Strike
- LockBit Ransomware
- AcidRain Malware
- Mis-Type Malware

Malware analysis includes studying execution methods, persistence mechanisms, command-and-control communication, credential theft, and exfiltration behavior.

6. Indicators of Compromise (IoCs)

Indicators of Compromise (IoCs) are forensic artifacts indicating malicious activity. Common IoCs include:

- Suspicious IP addresses
- Malicious domains and URLs
- File hashes (MD5/SHA256)
- Registry changes
- Phishing email attachments
- PowerShell execution logs
- Abnormal outbound traffic

7. MITRE ATT&CK; Technique Mapping

Common ATT&CK; techniques observed during threat monitoring include:

- T1059.001 – PowerShell
- T1566.001 – Spearphishing Attachment
- T1003 – Credential Dumping
- T1547 – Boot or Logon Autostart Execution
- T1071.001 – Web Protocols
- T1055 – Process Injection
- T1041 – Exfiltration Over C2 Channel

8. Threat Hunting & Detection Engineering

Threat hunting involves proactively searching systems and logs for malicious indicators. Detection engineering focuses on creating SIEM, EDR, and SOC detection rules. Security analysts monitor suspicious PowerShell activity, unauthorized login attempts, abnormal process execution, and unusual outbound network traffic.

9. Ransomware Operations

Modern ransomware groups use phishing campaigns, stolen credentials, exploit kits, and remote access tools to compromise victims. Ransomware operators often exfiltrate sensitive data before encryption to pressure victims into paying ransom demands.

10. Operational Security Recommendations

Organizations should implement the following security measures:

- Multi-factor authentication (MFA)
- Endpoint Detection & Response (EDR)
- Security awareness training
- Network segmentation
- Threat intelligence integration
- Regular vulnerability patching

- Secure email filtering
- Continuous log monitoring

11. Conclusion

This research improved understanding of cyber threat intelligence, malware behavior, ATT&CK; mapping, IoC collection, and APT monitoring operations. Continuous monitoring, defensive analysis, and structured reporting are essential for modern cybersecurity operations and incident response.

Threat Intelligence Monitoring Table

Threat Group	Primary Objective	Common Techniques
Lazarus Group	Financial Theft	T1566.001, T1059
APT29	Cyber Espionage	T1003, T1071
Sandworm Team	Infrastructure Attacks	T1059.001
Gallmaker	Military Espionage	T1547, T1071
LockBit	Ransomware Operations	T1486, T1041

12. References

- <https://attack.mitre.org/>
- <https://attack.mitre.org/groups/>
- <https://attack.mitre.org/software/>
- <https://attack.mitre.org/techniques/enterprise/>